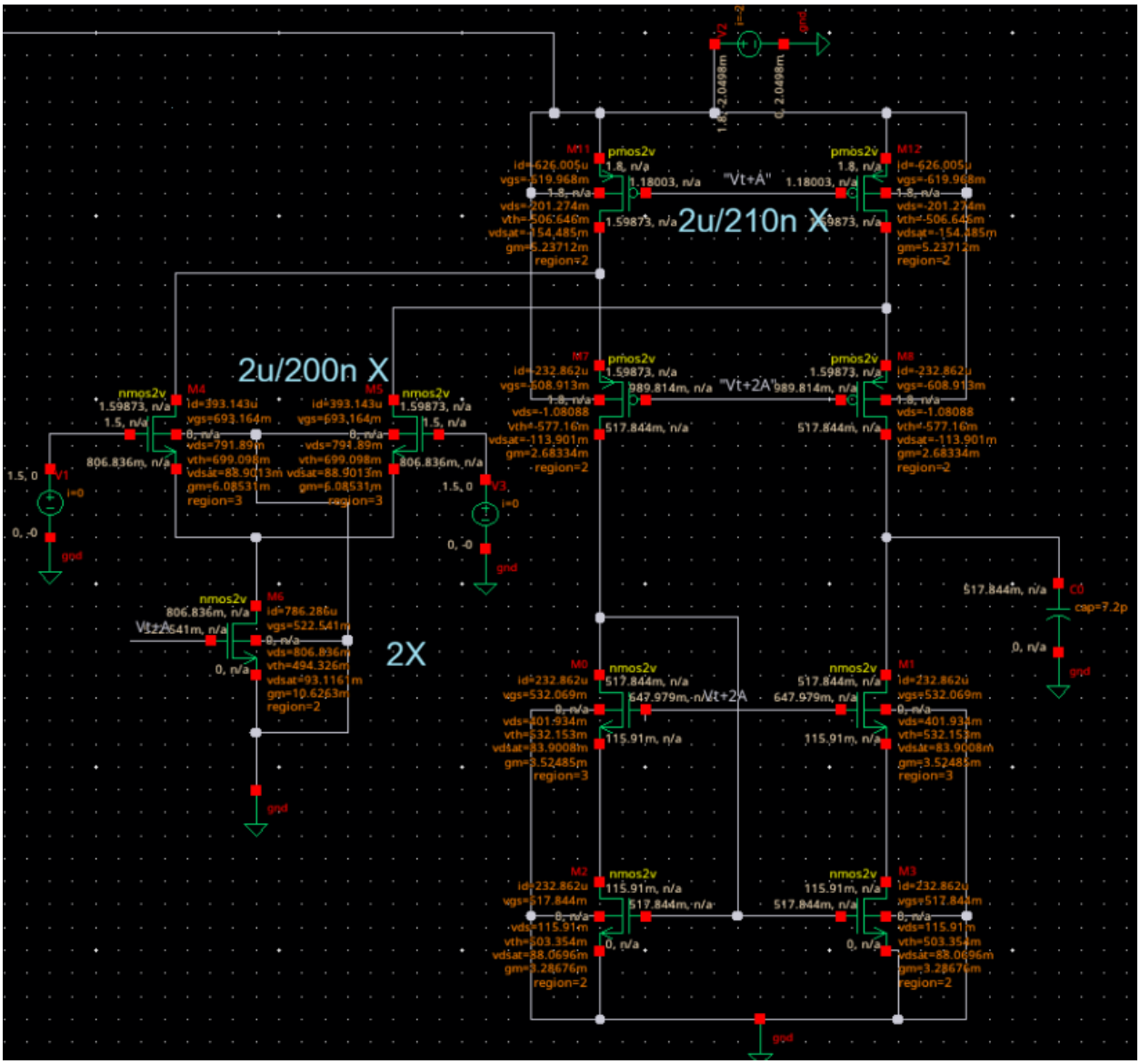
Differential Input Singled-Ended Output Folded-Cascode Operational Amplifier

<u>Folded Cascode Circuit</u>



Variables

V_{cm}	V_{supply}	I	L	W	C_{load}
$1.5V$	$1.8V$	$45\mu A$	$180nm$ (Unless specified otherwise)	$45\mu m$	$6pF \pm 1.2pF$

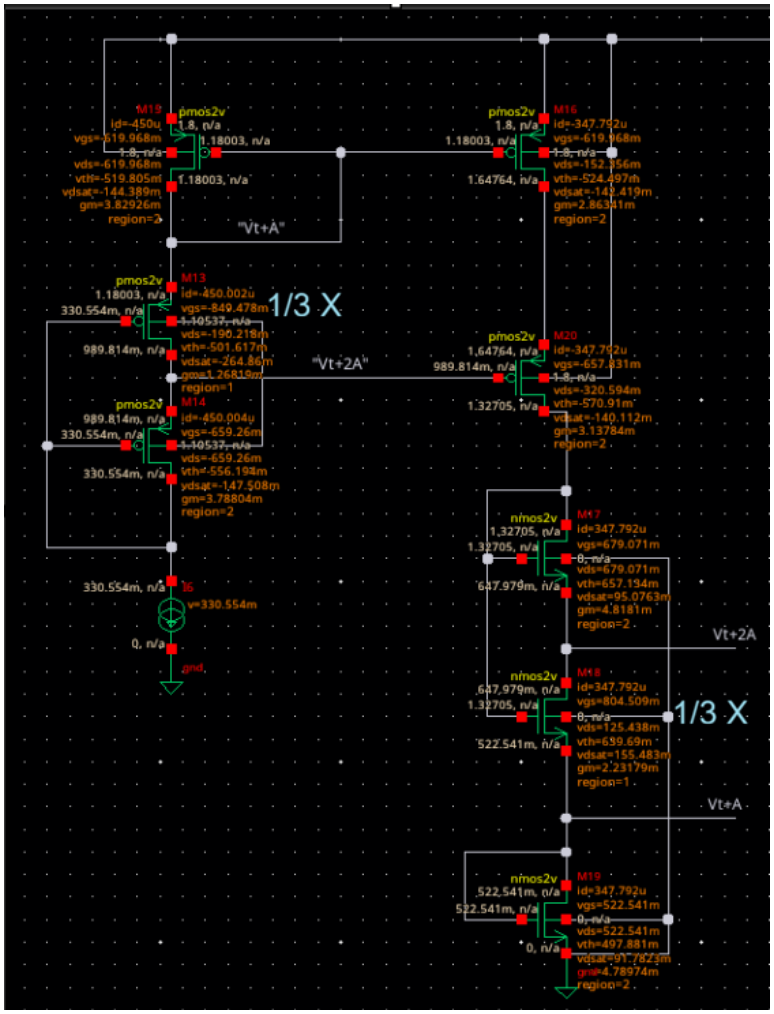
Desired Specs

DC gain	>50dB
Unity-gain BW	>200MHz
PM	>60°
Output swing	>0.5 Vpp
Power consumption	<12 mW
Input-referred systematic offset	<3mV

Achieved Specs

	Typical	Fast	Slow
DC gain	52.57dB	51.57dB	53.2dB
Unity-gain BW	284.1 MHz	357MHz	227.2 MHz
PM	69°	66.6°	68.5°
Output swing	>0.5 Vpp	>0.5 Vpp	>0.5 Vpp
Power consumption	6mW	6.77 mW	5.8mW
Input-referred systematic offset	2.362mV	2.391mV	2.442mV

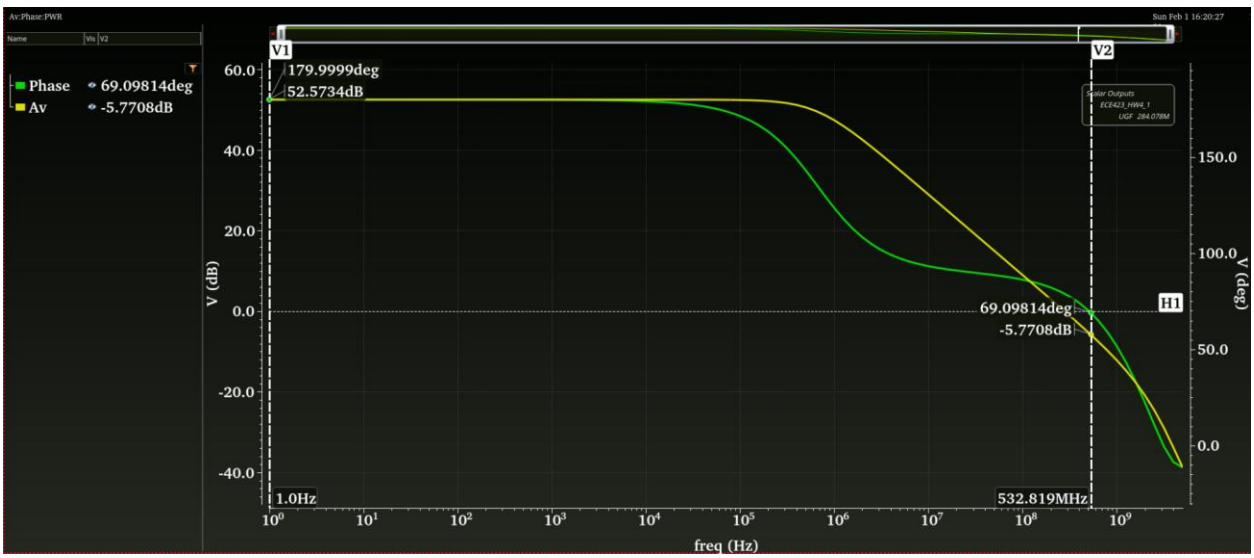
Sooch Biasing circuit



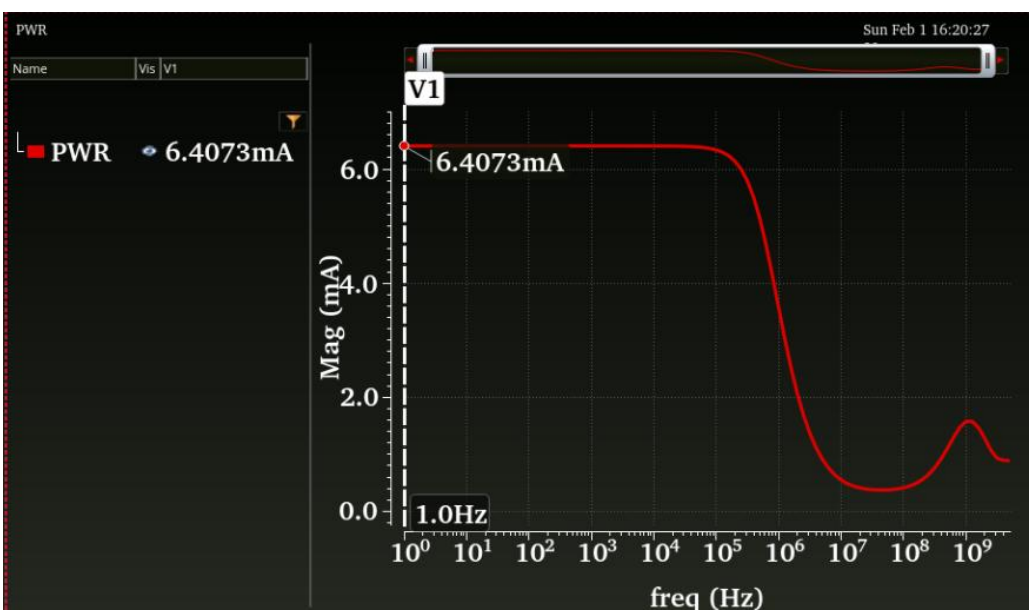
Normal (temp = 27°C, C_{load} = 6pF)

Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
ECE423_HW4_1	Av				
ECE423_HW4_1	UGF	284.1M	> 200M		pass
ECE423_HW4_1	Phase				
ECE423_HW4_1	PWR				

Phase & Gain Plot (PM = $69°$, $Av = 52.57\ dB$)



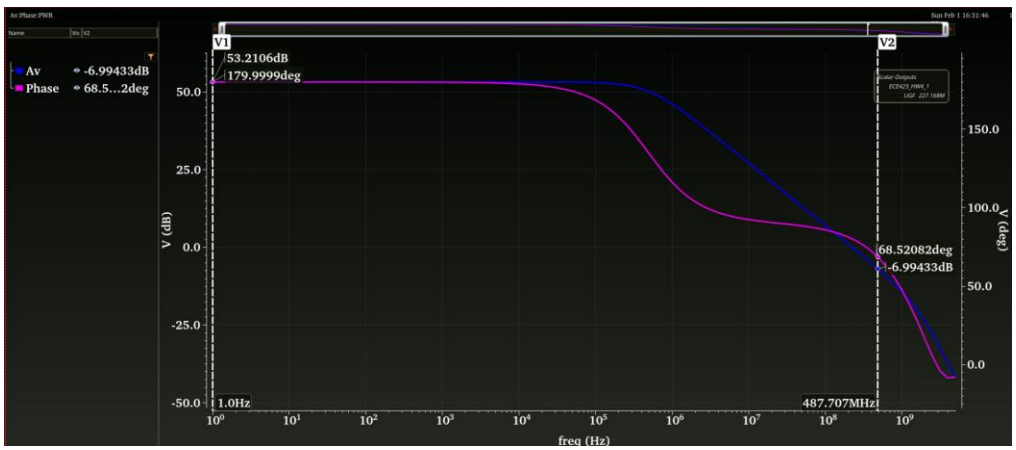
Power Plot (PWR_max = ~ 6mW)



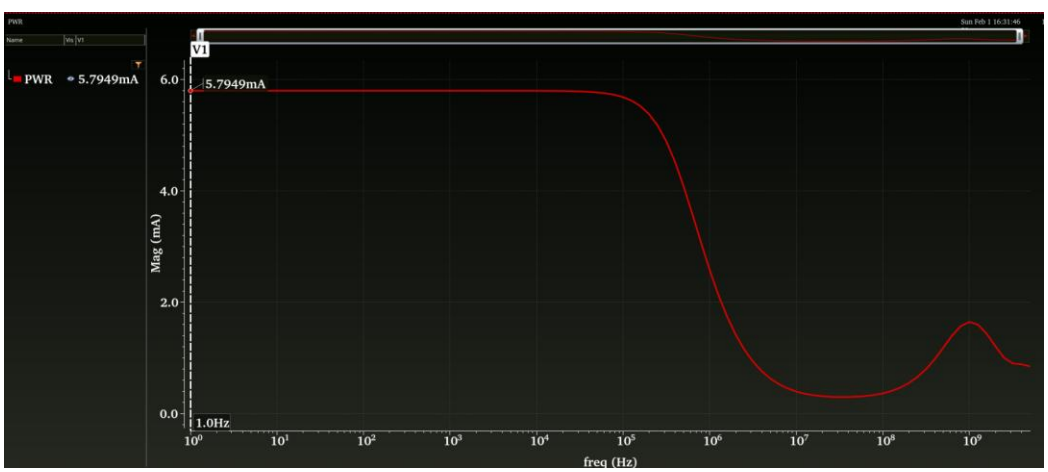
Slow $(temp = 100°C, C_{load} = 7.2pF)$

Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
ECE423_HW4_1	Av				
ECE423_HW4_1	UGF	227.2M	> 200M		pass
ECE423_HW4_1	Phase				
ECE423_HW4_1	PWR				

Phase & Gain Plot (PM = 68.5°, Av = 53.2 dB)



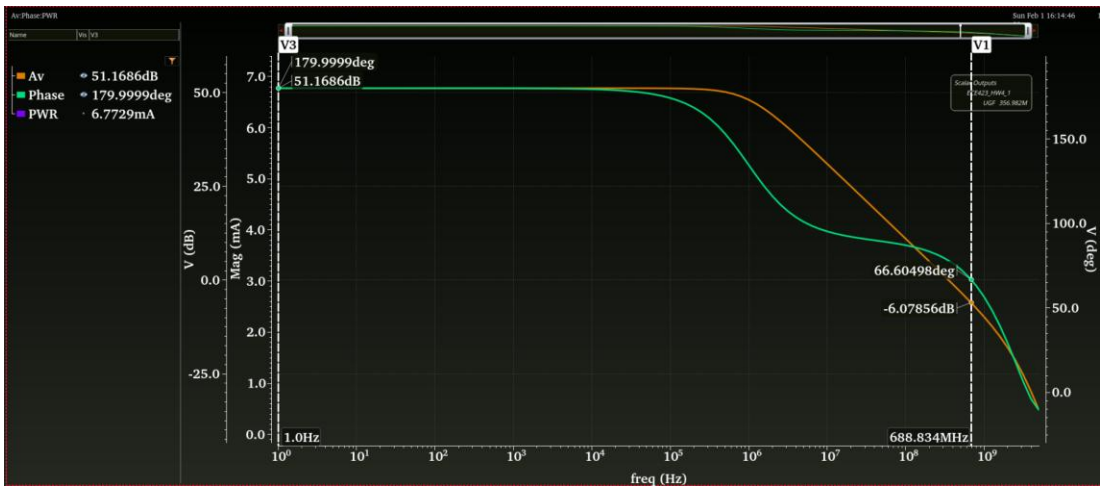
Power Plot (PWR_max = 5.8 mW)



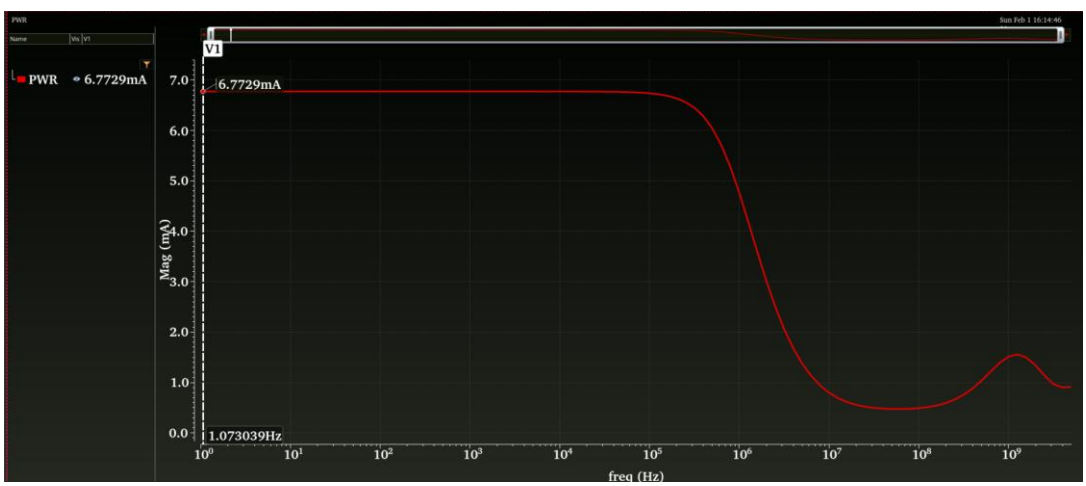
$Fast\,(temp\ =\ -40°C,\ C_{load} = 4.8pF$

Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
ECE423_HW4_1	Av				
ECE423_HW4_1	UGF	357M	> 200M		pass
ECE423_HW4_1	Phase				
ECE423_HW4_1	PWR				

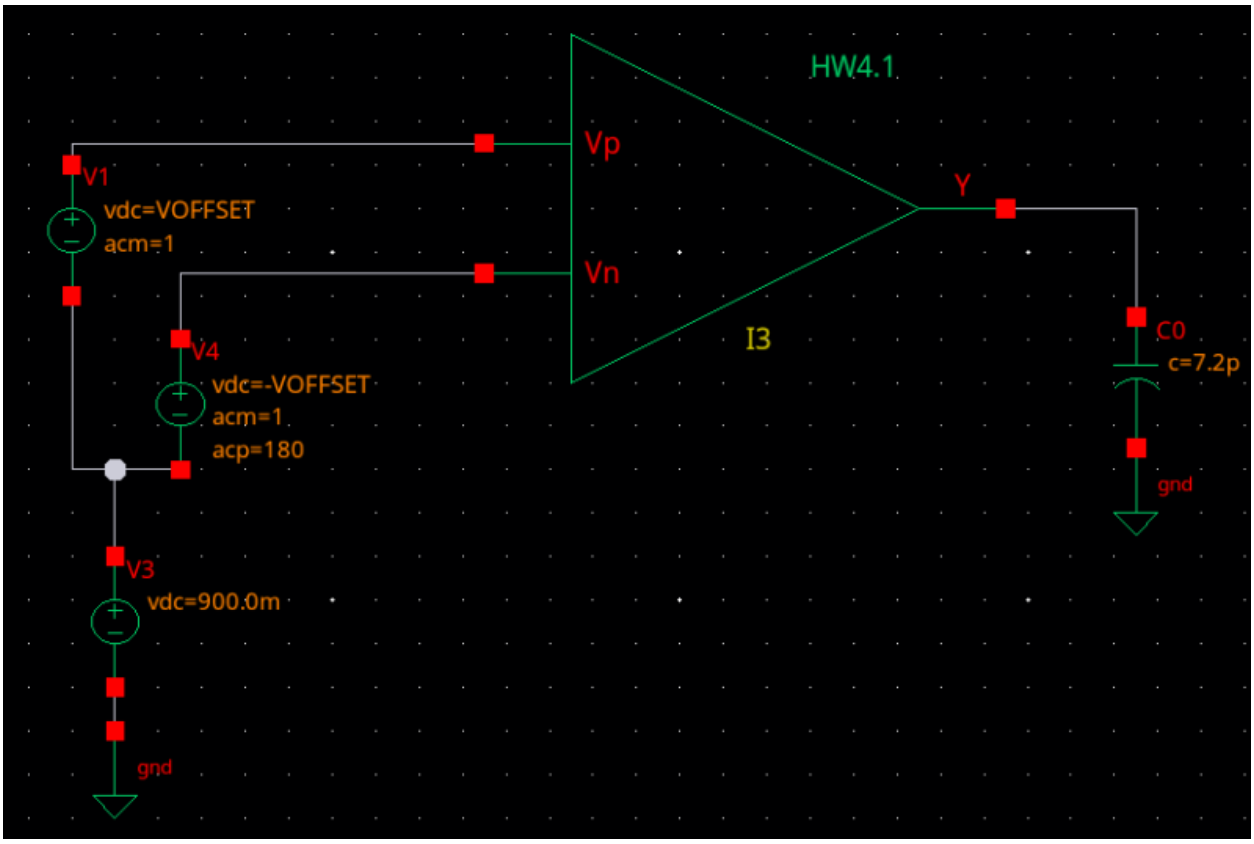
Phase & Gain Plot (PM = 66.6°, Av = 51.17 dB)



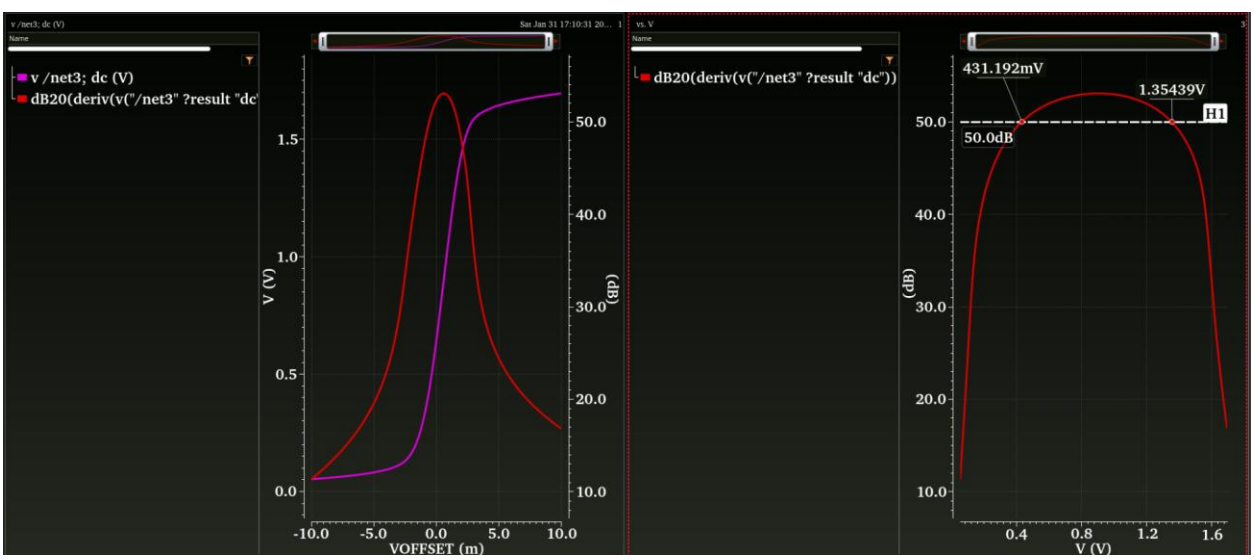
Power Plot (PWR_max = 6.77 mW)



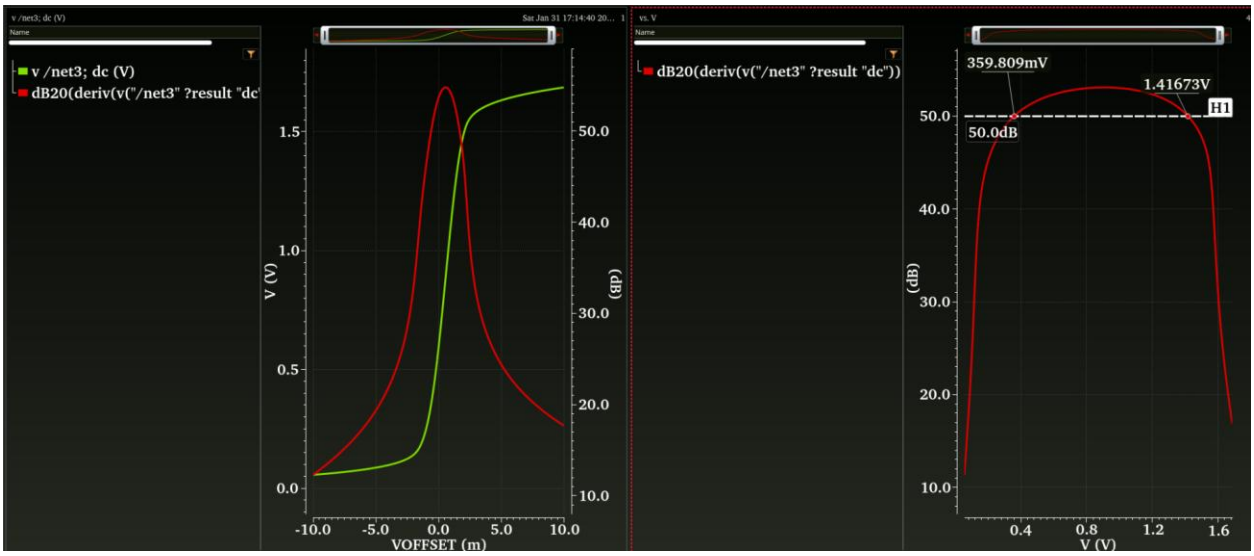
Voltage Swing:



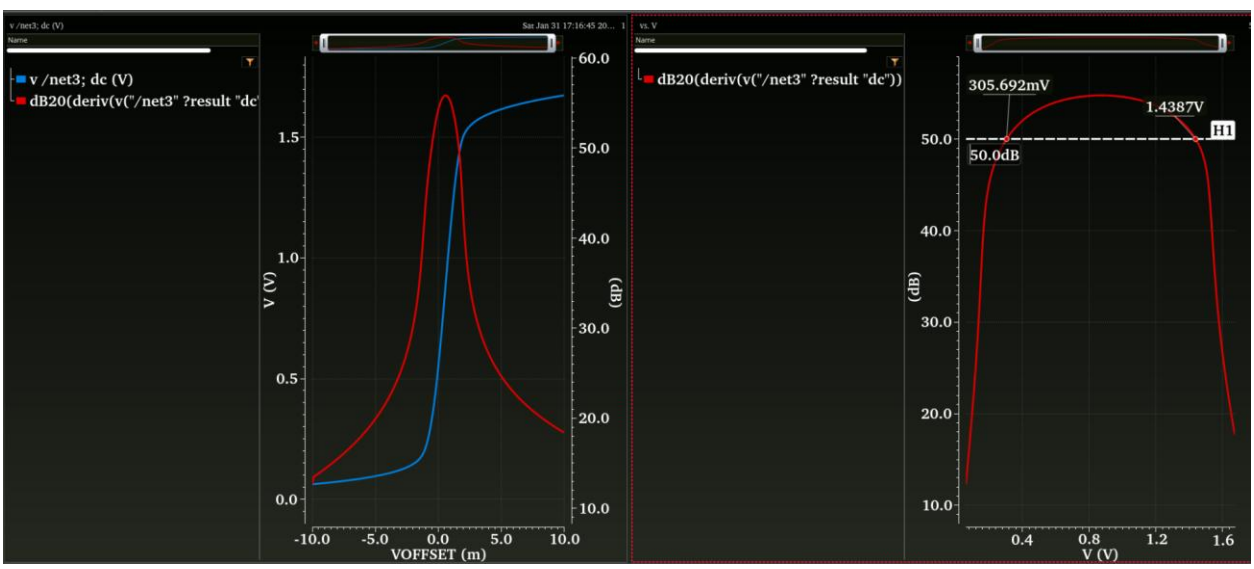
Temperature $= -40°C$ *and Capacitor* $= 4.8pF$



Temperature $= \mathbf{27°C}$ *and Capacitor = 6pF*



Temperature $= \mathbf{100°C}$ *and Capacitor = 7.2pF*

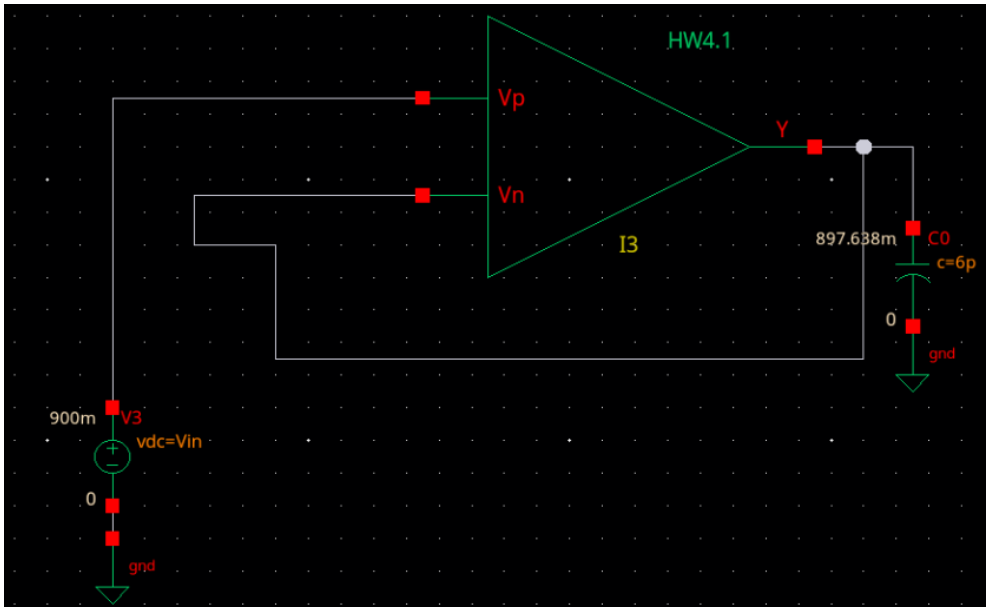


Systemic Offset

Normal (Temp = 27°C, $\boldsymbol{C_{load} = 6pF}$*)*

Vn = 897.638mV ≅ 900mV

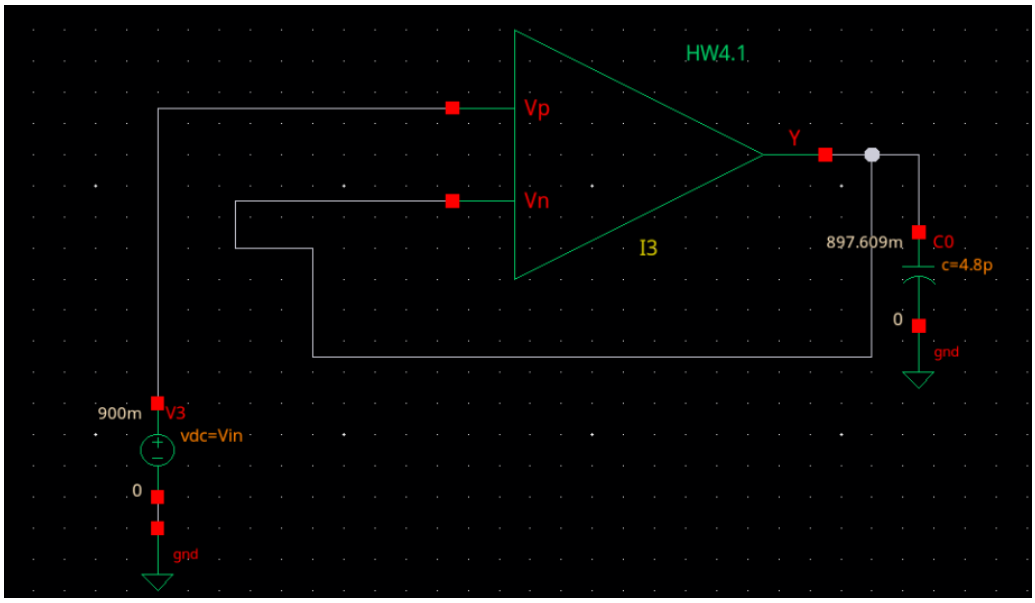
Systemic offset= $\boldsymbol{2.362\ mV}$



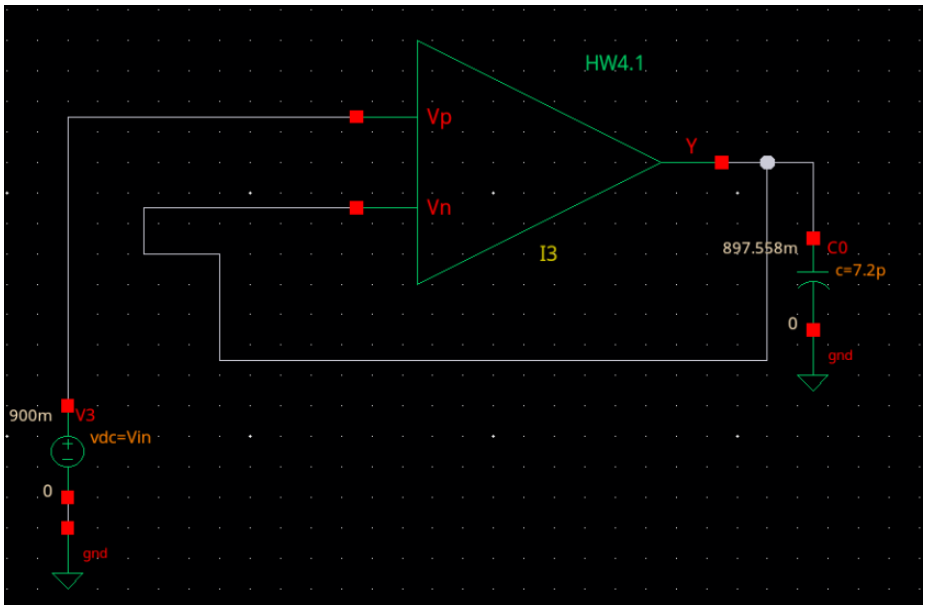
Fast (Temp = -40°C, $\boldsymbol{C_{load} = 4.8pF}$*)*

Vn = 897.609mV ≅ 900mV

Systemic offset ≅ $\boldsymbol{2.391mV}$



Slow (Temp = 100°C, $\boldsymbol{C_{load} = 7.2pF}$*)*

Vn = 897.6mV ≅ 900mV

Systemic offset = $\boldsymbol{2.442\ mV}$